一 不定积分的性质

(1) $\int [f(x) \pm g(x)]dx = \int f(x)dx \pm \int g(x)dx;$

证 $\because \left[\int f(x)dx \pm \int g(x)dx\right]'$

$= \left[\int f(x)dx\right]' \pm \left[\int g(x)dx\right]' = f(x) \pm g(x).$

$\therefore$ 等式成立.

（此性质可推广到有限多个函数之和的情况）

(2) $\int kf(x)dx = k\int f(x)dx.$

（k 是常数，$k \neq 0$)

例1 求积分 $\int(\frac{3}{1+x^2} - \frac{2}{\sqrt{1-x^2}})dx.$

解 $\int(\frac{3}{1+x^2} - \frac{2}{\sqrt{1-x^2}})dx$

$$= 3\int\frac{1}{1+x^2}dx - 2\int\frac{1}{\sqrt{1-x^2}}dx$$

$$= 3\arctan x - 2\arcsin x + C$$

例2 求积分 $\int \frac{1+x+x^2}{x(1+x^2)}dx.$

解 $$\int \frac{1+x+x^2}{x(1+x^2)}dx = \int \frac{x+(1+x^2)}{x(1+x^2)}dx$$

$$= \int \left(\frac{1}{1+x^2} + \frac{1}{x} \right) dx = \int \frac{1}{1+x^2}dx + \int \frac{1}{x}dx$$

$$= \arctan x + \ln|x| + C.$$

例3 求积分 $\int \frac{1+2x^2}{x^2(1+x^2)}dx.$

解 $\int \frac{1+2x^2}{x^2(1+x^2)}dx = \int \frac{1+x^2+x^2}{x^2(1+x^2)}dx$

$$= \int \frac{1}{x^2}dx + \int \frac{1}{1+x^2}dx$$

$$= -\frac{1}{x} + \arctan x + C.$$

例4 求积分 $\int \frac{1}{1+\cos 2x} dx.$

解 $$\int \frac{1}{1+\cos 2x} dx = \int \frac{1}{1+2\cos^2 x - 1} dx$$

$$= \frac{1}{2}\int \frac{1}{\cos^2 x} dx = \frac{1}{2}\tan x + C.$$

说明： 以上几例中的被积函数都需要进行恒等变形，才能使用基本积分表.

例 5 已知一曲线$y=f(x)$在点$(x,f(x))$处的切线斜率为$\sec^2 x+\sin x$，且此曲线与y轴的交点为$(0,5)$，求此曲线的方程.

解 $\because \quad \dfrac{dy}{dx}=\sec^2 x+\sin x,$

$$\therefore \quad y=\int\left(\sec^2 x+\sin x\right)dx$$

$$=\tan x-\cos x+C,$$

$\because \quad y(0)=5, \qquad \therefore \quad C=6,$

所求曲线方程为 $y=\tan x-\cos x+6.$

思考与练习

1. 若 e^{-x} 是 $f(x)$的原函数，则

$$\int x^2 f(\ln x)\,\mathrm{d}x = \underline{\quad -\frac{1}{2}x^2 + C \quad}$$

提示：$f(x) = (e^{-x})' = -e^{-x}$

$$f(\ln x) = -e^{-\ln x} = -\frac{1}{x}$$

2. 若 $f(x)$ 的导函数为 $\sin x$,则 $f(x)$ 的一个原函数是(B).

(A) $1+\sin x$; (B) $1-\sin x$;

(C) $1+\cos x$; (D) $1-\cos x$.

提示: 已知 $f'(x)=\sin x$

求 $(?)'=f(x)$

即 $(?)''=\sin x$

或由题意 $f(x)=-\cos x+C_1$, 其原函数为

$$\int f(x)\,\mathrm{d}x=-\sin x+C_1x+C_2$$

在一般情况下：

设 $F'(u) = f(u)$, 则 $\int f(u)du = F(u) + C$.

如果 $u = \varphi(x)$（可微）

$$\because \quad dF[\varphi(x)] = f[\varphi(x)]\varphi'(x)dx$$

$$\therefore \quad \int f[\varphi(x)]\varphi'(x)dx = F[\varphi(x)] + C$$

$$= [\int f(u)du]_{u=\varphi(x)}$$ 由此可得换元法定理

二 不定积分的第一换元法（凑微分）

定理1 如果$u=g(x)$是可微函数，$f(u)$在$g(x)$的值域区间I上连续，则

$$\int f(g(x))g'(x)dx=\int f(g(x))d(g(x))$$

$$\overset{u=g(x)}{=}\int f(u)du$$

例　求 $\int \sin 2x dx.$

解（一）$\int \sin 2x dx = \frac{1}{2}\int \sin 2x (2x)' dx = \frac{1}{2}\int \sin 2x d(2x)$

$$= -\frac{1}{2}\cos 2x + C;$$

解（二）$\int \sin 2x dx = 2\int \sin x \cos x dx$

$$= 2\int \sin x d(\sin x) = (\sin x)^2 + C;$$

解（三）$\int \sin 2x dx = 2\int \sin x \cos x dx$

$$= -2\int \cos x d(\cos x) = -(\cos x)^2 + C.$$

例　求 $\int \frac{1}{3+2x}dx.$

解　$\frac{1}{3+2x}=\frac{1}{2}\cdot\frac{1}{3+2x}\cdot(3+2x)',$

$$\int \frac{1}{3+2x}dx=\frac{1}{2}\int \frac{1}{3+2x}\cdot(3+2x)'dx$$

$$=\frac{1}{2}\int \frac{1}{u}du=\frac{1}{2}\ln|u|+C=\frac{1}{2}\ln|3+2x|+C.$$

一般地　$\int f(ax+b)dx=\frac{1}{a}[\int f(u)du]_{u=ax+b}$

例　求 $\int \frac{1}{x(1+2\ln x)}dx.$

解　$\int \frac{1}{x(1+2\ln x)}dx = \int \frac{1}{1+2\ln x}d(\ln x)$

$$= \frac{1}{2}\int \frac{1}{1+2\ln x}d(1+2\ln x)$$

$$u = 1+2\ln x$$

$$= \frac{1}{2}\int \frac{1}{u}du = \frac{1}{2}\ln u + C = \frac{1}{2}\ln(1+2\ln x) + C.$$

例　求 $\int \frac{x}{(1+x)^3}dx.$

解　$\int \frac{x}{(1+x)^3}dx = \int \frac{x+1-1}{(1+x)^3}dx$

$$= \int [\frac{1}{(1+x)^2} - \frac{1}{(1+x)^3}]d(1+x)$$

$$= -\frac{1}{1+x} + C_1 + \frac{1}{2(1+x)^2} + C_2$$

$$= -\frac{1}{1+x} + \frac{1}{2(1+x)^2} + C.$$

例 求 $\int \frac{1}{a^2+x^2}dx.$

解 $\int \frac{1}{a^2+x^2}dx = \frac{1}{a^2}\int \frac{1}{1+\frac{x^2}{a^2}}dx$

$$= \frac{1}{a}\int \frac{1}{1+\left(\frac{x}{a}\right)^2}d\left(\frac{x}{a}\right) = \frac{1}{a}\arctan\frac{x}{a}+C.$$

例　求 $\int \frac{1}{x^2-8x+25}dx.$

解 $\int \frac{1}{x^2-8x+25}dx = \int \frac{1}{(x-4)^2+9}dx$

$$= \frac{1}{3^2}\int \frac{1}{\left(\frac{x-4}{3}\right)^2+1}dx = \frac{1}{3}\int \frac{1}{\left(\frac{x-4}{3}\right)^2+1}d\left(\frac{x-4}{3}\right)$$

$$= \frac{1}{3}\arctan\frac{x-4}{3}+C.$$

例. $\int \frac{6}{x^2-4x-5}dx.$

解: $\frac{6}{x^2-4x-5}=\frac{6}{(x-5)(x+1)}=\frac{A}{x-5}+\frac{B}{x+1}$

$\Rightarrow\ 6=A(x+1)+B(x-5),$

令$x=-1,\ \Rightarrow B=-1,$ 令$x=5,\ \Rightarrow A=1,$

$$\int \frac{6}{x^2-4x-5}dx=\int\left(\frac{1}{x-5}+\frac{-1}{x+1}\right)dx$$

$$=\ln|x-5|-\ln|x+1|+C$$

例　求　$\int \frac{1}{1+e^x}dx.$

解　$\int \frac{1}{1+e^x}dx = \int \frac{1+e^x-e^x}{1+e^x}dx$

$$= \int \left(1-\frac{e^x}{1+e^x}\right)dx = \int dx - \int \frac{e^x}{1+e^x}dx$$

$$= \int dx - \int \frac{1}{1+e^x}d(1+e^x)$$

$$= x - \ln(1+e^x) + C.$$

例　求　$\int (1-\frac{1}{x^2})e^{x+\frac{1}{x}}dx.$

解　$\because \left(x+\frac{1}{x}\right)' = 1-\frac{1}{x^2},$

$$\therefore \int (1-\frac{1}{x^2})e^{x+\frac{1}{x}}dx$$

$$= \int e^{x+\frac{1}{x}}d(x+\frac{1}{x}) = e^{x+\frac{1}{x}} + C.$$

例 求 $\int \frac{1}{\sqrt{2x+3}+\sqrt{2x-1}}dx.$

原式$=\int \frac{\sqrt{2x+3}-\sqrt{2x-1}}{\left(\sqrt{2x+3}+\sqrt{2x-1}\right)\left(\sqrt{2x+3}-\sqrt{2x-1}\right)}dx$

$$=\frac{1}{4}\int \sqrt{2x+3}dx-\frac{1}{4}\int \sqrt{2x-1}dx$$

$$=\frac{1}{8}\int \sqrt{2x+3}d(2x+3)-\frac{1}{8}\int \sqrt{2x-1}d(2x-1)$$

$$=\frac{1}{12}\left(\sqrt{2x+3}\right)^3-\frac{1}{12}\left(\sqrt{2x-1}\right)^3+C.$$

一般地

$$(1)\ \int f(ax+b)dx=\frac{1}{a}\int f(ax+b)d(ax+b)$$

$$=\frac{1}{a}[\int f(u)du]_{u=ax+b}$$

$$(2)\ \int xf(ax^2+b)dx=\frac{1}{2a}\int f(ax^2+b)d(ax^2+b)$$

$$(3)\int x^{\alpha}f(ax^{\alpha+1}+b)dx$$

$$=\frac{1}{a(\alpha+1)}\int f(ax^{\alpha+1}+b)d(ax^{\alpha+1}+b)$$

例 求 $\int \frac{1}{1+\cos x}dx.$

解 $\int \frac{1}{1+\cos x}dx = \int \frac{1-\cos x}{(1+\cos x)(1-\cos x)}dx$

$$= \int \frac{1-\cos x}{1-\cos^2 x}dx = \int \frac{1-\cos x}{\sin^2 x}dx$$

$$= \int \frac{1}{\sin^2 x}dx - \int \frac{1}{\sin^2 x}d(\sin x)$$

$$= -\cot x + \frac{1}{\sin x} + C.$$

总结(1)$\int \sin^{2m+1} x dx = \int \sin^{2m} x \sin x dx$

$$= -\int (1-\cos^2 x)^m d(\cos x)$$

(2)$\int \cos^{2m+1} x dx = \int (1-\sin^2 x)^m d(\sin x)$

(3)$\int \sin^{2m} x dx, \int \cos^{2m} x dx$ 利用倍角公式降幂

(4)$\int \sin \alpha x \cos \beta x dx, \int \sin \alpha x \sin \beta x dx, \int \cos \alpha x \cos \beta x dx$

利用积化和差公式

例　求　$\int \cos 3x \cos 2x dx.$

解　$\cos A \cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)],$

$$\cos 3x \cos 2x = \frac{1}{2}(\cos x + \cos 5x),$$

$$\int \cos 3x \cos 2x dx = \frac{1}{2}\int (\cos x + \cos 5x) dx$$

$$= \frac{1}{2}\sin x + \frac{1}{10}\sin 5x + C.$$

例 求$\int \sin^2 x \cdot \cos^5 x dx$.

解 $\int \sin^2 x \cdot \cos^5 x dx = \int \sin^2 x \cdot \cos^4 x d(\sin x)$

$$= \int \sin^2 x \cdot (1 - \sin^2 x)^2 d(\sin x)$$

$$= \int (\sin^2 x - 2\sin^4 x + \sin^6 x) d(\sin x)$$

$$= \frac{1}{3}\sin^3 x - \frac{2}{5}\sin^5 x + \frac{1}{7}\sin^7 x + C.$$

说明 当被积函数是三角函数相乘时，拆开奇次项去凑微分.

$$(1)\int \sin^{2m+1} x\cos^{2m} x dx = \int \sin^{2m} x\cos^{2m} x\sin x dx$$

$$= -\int (1-\cos^2 x)^m \cos^{2m} x d(\cos x)$$

$$(2)\int \cos^{2m+1} x\sin^{2m} x dx = \int \cos^{2m} x\sin^{2m} x\cos dx$$

$$= \int (1-\sin^2 x)^m \sin^{2m} x d(\sin x)$$

例 求 $\int \csc x dx.$

解（一）$\int \csc x dx = \int \frac{1}{\sin x} dx = \int \frac{1}{2\sin\frac{x}{2}\cos\frac{x}{2}} dx$

$$= \int \frac{1}{\tan\frac{x}{2}\left(\cos\frac{x}{2}\right)^2} d\left(\frac{x}{2}\right) = \int \frac{1}{\tan\frac{x}{2}} d\left(\tan\frac{x}{2}\right)$$

$$= \ln\left|\tan\frac{x}{2}\right| + C = \ln\left|\csc x - \cot x\right| + C.$$

（使用了三角函数恒等变形）

解（二）$\int \csc x dx = \int \frac{1}{\sin x} dx = \int \frac{\sin x}{\sin^2 x} dx$

$$= -\int \frac{1}{1-\cos^2 x} d(\cos x) \qquad u = \cos x$$

$$= -\int \frac{1}{1-u^2} du = -\frac{1}{2}\int \left(\frac{1}{1-u} + \frac{1}{1+u} \right) du$$

$$= \frac{1}{2}\ln\left|\frac{1-u}{1+u}\right| + C = \frac{1}{2}\ln\left|\frac{1-\cos x}{1+\cos x}\right| + C.$$

类似地可推出 $\int \sec x dx = \ln\left|\sec x + \tan x\right| + C.$

例 设 $f'(\sin^2 x)=\cos^2 x$,求 $f(x)$.

解 令 $u=\sin^2 x \Rightarrow \cos^2 x=1-u$,

$$f'(u)=1-u,$$

$$f(u)=\int(1-u)du=u-\frac{1}{2}u^2+C,$$

$$f(x)=x-\frac{1}{2}x^2+C.$$

例 求 $\int \frac{1}{\sqrt{4-x^2}\arcsin\frac{x}{2}}dx.$

解 $\int \frac{1}{\sqrt{4-x^2}\arcsin\frac{x}{2}}dx = \int \frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}\arcsin\frac{x}{2}}d\frac{x}{2}$

$$= \int \frac{1}{\arcsin\frac{x}{2}}d\left(\arcsin\frac{x}{2}\right) = \ln\left|\arcsin\frac{x}{2}\right| + C.$$

要熟记常见的微分式

$$xdx = \frac{1}{2}d(x^2), \qquad \frac{1}{ax+b}dx = \frac{1}{a}\ln(ax+b),(ax+b>0)$$

$$\frac{1}{\sqrt{x}}dx = 2d(\sqrt{x}), \quad \frac{1}{x^2}dx = -d(\frac{1}{x}), \qquad e^x d = d(e^x),$$

$$\sin xdx = -d(\cos x), \qquad \cos xdx = d(\sin x),$$ 等

例.求 $\int \sec x dx$.

解: $\int \sec x dx = \int \frac{1}{\cos x} dx = \int \frac{\cos x}{\cos^2 x} dx$

$$= \int \frac{1}{1-\sin^2 x} d\sin x = -\int \frac{1}{\sin^2 x - 1} d\sin x$$

$$= -\frac{1}{2}\ln\left|\frac{\sin x - 1}{\sin x + 1}\right| + C = \frac{1}{2}\ln\left|\frac{\sin x + 1}{\sin x - 1}\right| + C$$

$$= \frac{1}{2}\ln\frac{(\sin x + 1)^2}{\cos^2 x} + C = \ln\left|\frac{\sin x + 1}{\cos x}\right| + C$$

$$= \ln|\sec x + \tan x| + C.$$

同理: $\int \csc x dx = \ln|\csc x - \cot x| + C.$

例求 $\int \sec x dx.$

法二: $\int \sec x dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} dx$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx = \int \frac{1}{\sec x + \tan x} d(\sec x + \tan x)$$

$$= \ln|\sec x + \tan x| + C.$$

例. $\int \frac{x+7}{x^2-4x-5}dx.$

解: $\frac{x+7}{x^2-4x-5} = \frac{x+7}{(x-5)(x+1)} = \frac{A}{x-5} + \frac{B}{x+1}$

$\Rightarrow \quad x+7 = A(x+1) + B(x-5),$

令$x=-1,\ \Rightarrow B=-1,$ 令$x=5,\ \Rightarrow A=2,$

$$\int \frac{x+7}{x^2-4x-5}dx = \int \left(\frac{2}{x-5} + \frac{-1}{x+1}\right)dx$$

$$= 2\ln|x-5| - \ln|x+1| + C$$

例. $\int \frac{x-2}{x^2+2x+3}dx.$

解: $\int \frac{x-2}{x^2+2x+3}dx = \int \frac{\frac{1}{2}(2x+2)-3}{x^2+2x+3}dx$

$$= \frac{1}{2}\int \frac{2x+2}{x^2+2x+3}dx - 3\int \frac{dx}{(x+1)^2+(\sqrt{2})^2}$$

$$= \frac{1}{2}\int \frac{1}{x^2+2x+3}d(x^2+2x+3) - 3\int \frac{d(x+1)}{(x+1)^2+(\sqrt{2})^2}$$

$$= \frac{1}{2}\ln(x^2+2x+3) - \frac{3}{\sqrt{2}}\arctan\frac{x+1}{\sqrt{2}} + C.$$

思考题

已知 $\frac{\cos x}{x}$ 是 $f(x)$ 的一个原函数，则

$$\int f(x)\frac{\cos x}{x}dx = \underline{\frac{1}{2}\left(\frac{\cos x}{x}\right)^2 + C}$$

解：$\because d\left(\frac{\cos x}{x}\right) = f(x)dx$

$$\therefore \int f(x)\frac{\cos x}{x}dx = \int \frac{\cos x}{x}d\left(\frac{\cos x}{x}\right) = \frac{1}{2}\left(\frac{\cos x}{x}\right)^2 + C$$

不定积分 $\int \frac{1}{\sqrt{x(1-x)}}dx =$ （ D

（A） $\frac{1}{2}\arcsin\sqrt{x}+C$ ；

（B） $\arcsin\sqrt{x}+C$ ；

（C） $2\arcsin(2x-1)+C$ ；

（D） $\arcsin(2x-1)+C$.

解： $\int \frac{1}{\sqrt{x(1-x)}}dx = \int \frac{1}{\sqrt{\frac{1}{4}-\left(x-\frac{1}{2}\right)^2}}dx = 2\int \frac{1}{\sqrt{1-(2x-1)^2}}dx$

$$= \int \frac{1}{\sqrt{1-(2x-1)^2}}d(2x-1) = \arcsin(2x-1)+C$$